

RIK JANSEN

Senior Software Engineer

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SUMMARY

Senior AI/ML Engineer with 16+ years of software engineering experience across SaaS, healthcare, fintech, logistics, and enterprise automation, specializing in production-grade machine learning platforms using **Python 3.8–3.12**, **PyTorch 1.12–2.x**, **TensorFlow 2.x**, **FastAPI 0.95–0.110**, **MLflow 2.x**, **Airflow 2.x**, **Docker 20–25**, **Kubernetes 1.23–1.29**, **AWS/GCP** cloud services, vector search, LLM integration, and data pipelines, with strong delivery in model deployment, API design, observability, migration, bug resolution, cost optimization, and measurable improvements in accuracy, latency, reliability, security compliance, and developer productivity across cross-functional product teams.

SKILLS

- **AI / Machine Learning:** **PyTorch 1.12–2.x**, **TensorFlow 2.x**, **scikit-learn 0.24–1.4**, **XGBoost 1.x–2.x**, **LightGBM 3.x–4.x**, **Hugging Face Transformers 4.x**, **LangChain 0.1–0.2**, **OpenAI API**, **RAG**, embeddings, classification, forecasting, anomaly detection, recommendation systems, model evaluation, feature engineering.
- **Programming:** **Python 3.8–3.12**, **TypeScript 4.x–5.x**, **JavaScript ES6+**, **SQL**, **Bash**, **REST API** design, **async** programming, data validation, background jobs.
- **Data Engineering:** **Pandas 1.x–2.x**, **NumPy 1.x–2.x**, **Apache Airflow 2.x**, **Spark 3.x**, **Kafka 2.x–3.x**, **dbt 1.x**, **PostgreSQL 12–16**, **BigQuery**, **Snowflake**, **Redis 6.x–7.x**, **Elasticsearch 7.x–8.x**.
- **MLOps / Deployment:** **MLflow 2.x**, **DVC 2.x–3.x**, **Docker 20–25**, **Kubernetes 1.23–1.29**, **Helm 3.x**, **GitHub Actions**, **GitLab CI**, **Jenkins**, model registry, **CI/CD** for ML, canary release, drift monitoring, rollback strategy.
- **Backend / APIs:** **FastAPI 0.95–0.110**, **Flask 1.x–2.x**, **Django 3.x–4.x**, **Node.js 14–20**, **GraphQL**, **OAuth2**, **JWT**, **OpenAPI**, microservices, distributed systems.
- **Cloud / DevOps:** **AWS SageMaker**, **AWS Lambda**, **ECS**, **EKS**, **S3**, **RDS**, **CloudWatch**, **GCP Vertex AI**, **Cloud Run**, **Cloud Functions**, **Pub/Sub**, **Terraform 1.x**, monitoring, logging, alerting.
- **Testing / Quality:** **Pytest 6.x–8.x**, **Great Expectations 0.15–0.18**, unit testing, integration testing, model validation, data quality checks, contract testing, performance profiling.

EXPERIENCE

Senior Software Engineer

Viaro Networks Inc.

June 2021 - October 2025

- Designed a **machine learning prediction platform** for SaaS operations using **Python 3.9–3.12**, **PyTorch 1.12–2.x**, **FastAPI 0.95–0.110**, and **MLflow 2.x**, enabling product teams to deploy scoring services with consistent versioning and rollback support.
- Built **real-time anomaly detection pipelines** with **Kafka 3.x**, **Redis 7.x**, **PostgreSQL 14–16**, and **Kubernetes 1.24–1.29**, reducing incident detection delay by **27%** across high-volume operational workflows.
- Migrated legacy batch-based scoring jobs from **Flask 1.x** and cron scripts into **FastAPI**, **Airflow 2.x**, and containerized services, improving model refresh reliability by **23%** while removing hidden manual deployment steps.
- Implemented **RAG-based document intelligence features** using **Hugging Face Transformers 4.x**, **OpenAI API**, vector embeddings, and **Elasticsearch 8.x**, allowing support teams to retrieve policy and ticket context faster.
- Resolved a production issue where inconsistent feature normalization caused model confidence drift, adding feature-store validation, **MLflow** comparison reports, and automated regression checks before release.
- Developed **OpenAPI-based ML inference endpoints** with **JWT** authentication, request validation, structured logging, and rate limiting, improving API stability for internal and partner-facing applications.
- Introduced **model monitoring dashboards** with **Prometheus**, **Grafana**, and custom **Python** metrics to track latency, prediction distribution, failed inference calls, and data quality degradation across environments.
- Optimized model serving containers by reducing unused dependencies, caching tokenizer assets, and tuning worker concurrency, cutting average inference latency by **19%** under peak traffic conditions.
- Collaborated with backend, data, and product teams to implement **role-based AI features**, ensuring sensitive predictions and audit explanations were visible only to authorized users.
- Improved **CI/CD** for ML models using **GitHub Actions**, **Docker 24–25**, **Kubernetes** deployments, and **MLflow** registry promotion rules, making staging-to-production releases more predictable and traceable.
- Fixed a memory leak in a long-running embedding service by profiling tensor lifecycle, batching vectorization calls, and clearing stale request objects after asynchronous processing completed.
- Created internal AI engineering guidelines for prompt safety, model versioning, dataset lineage, and rollback strategy, helping developers use **AI/ML tooling** more flexibly without increasing production risk.

Software Engineer

ArnAmy, Inc.

August 2017 - May 2021

- Developed **machine learning APIs** for customer analytics and workflow automation using **Python 3.7–3.9**, TensorFlow 2.x, scikit-learn 0.22–0.24, Flask 1.x–2.x, and PostgreSQL 11–13.
- Built classification and ranking models for business process prioritization, improving internal decision accuracy by **21%** through better feature engineering and cross-validation workflows.
- Migrated older Python 3.6 services to **Python 3.8–3.9** with dependency upgrades, test coverage improvements, and Docker-based runtime consistency across development and production environments.
- Implemented scheduled data pipelines with Airflow 1.10–2.x, Pandas 1.x, SQLAlchemy, and S3-compatible object storage to prepare training datasets from operational application data.
- Resolved an issue where duplicate records were inflating model metrics by adding data quality checks, deduplication logic, and validation reports before training jobs were executed.
- Integrated **TensorFlow 2.x model serving** behind Flask and REST APIs, allowing backend systems to consume predictions without direct dependency on training code or raw datasets.
- Improved slow recommendation queries by indexing high-cardinality fields, caching frequent lookups in Redis 5.x–6.x, and reducing average response time by **18%** for core workflows.
- Created model evaluation dashboards showing precision, recall, confusion matrix trends, and dataset coverage, helping product stakeholders understand prediction trade-offs before release.
- Worked on **NLP-based text classification** for support messages using TF-IDF, word embeddings, and early Transformer-based experiments where they were realistic for production usage.
- Implemented Docker-based CI pipelines with GitLab CI and Jenkins, ensuring Python tests, linting, API checks, and model validation completed before deployment approval.
- Refactored tightly coupled ML scripts into reusable services and shared Python modules, improving maintainability and making future feature delivery faster across multiple product teams.

Software Developer

Webiz Digital

December 2015 - July 2017

- Built data-driven backend services using **Python 3.5–3.6**, Django 1.11, Flask 0.12–1.x, scikit-learn 0.18–0.20, PostgreSQL 9.x–10.x, and JavaScript ES6 for digital platforms.
- Implemented early **customer segmentation models** with scikit-learn pipelines, clustering, and SQL-based feature extraction, helping marketing teams improve campaign targeting by **16%**.
- Migrated reporting logic from manual spreadsheet exports into automated Python and PostgreSQL workflows, reducing recurring data preparation time by **24%** for internal analysts.
- Created REST endpoints for analytics dashboards that exposed campaign performance, user behavior, and prediction outputs through secure backend services and structured JSON responses.
- Fixed a recurring issue where timezone mismatches caused incorrect daily metrics by normalizing timestamps in ETL scripts and adding database-level validation rules.
- Implemented basic recommendation logic using historical interaction data, weighted scoring, and SQL aggregation before later machine learning workflows became more standardized.
- Improved data import reliability by adding retry logic, error logging, malformed row handling, and clear failure reports for CSV and third-party API ingestion jobs.
- Worked with designers and product stakeholders to translate business questions into measurable data features, reports, and lightweight predictive models for client-facing products.
- Optimized slow analytics pages by rewriting inefficient joins, adding indexes, and caching repeated calculations, improving dashboard load speed by **20%** on large accounts.
- Contributed to code reviews and deployment preparation for Python services, ensuring model-related scripts followed consistent structure, testing, and operational documentation.

Junior Software Engineer

High Touch Technologies

August 2014 - November 2015

- Supported internal business applications using **Python 2.7–3.4**, JavaScript, SQL Server, MySQL, Flask 0.10, jQuery, and early data processing scripts for reporting workflows.
- Built automation scripts for cleaning operational data, detecting missing fields, and preparing weekly reports, reducing manual spreadsheet correction work by **15%** for support teams.

- Assisted senior engineers with early predictive reporting experiments using NumPy, Pandas 0.14–0.17, and scikit-learn 0.15–0.17 for basic classification and trend analysis.
- Resolved recurring import failures caused by inconsistent CSV encodings by adding validation, encoding fallback logic, and clearer error messages for non-technical users.
- Created SQL queries and stored procedures for operational dashboards, helping managers monitor workflow volume, ticket status, and recurring customer issues more efficiently.
- Improved legacy web forms by adding server-side validation, better error handling, and cleaner database constraints, reducing repeated data entry mistakes by **12%**.
- Worked on bug fixes across backend and frontend components, learning how production defects moved from user reports through debugging, patching, testing, and deployment.
- Documented setup steps, common errors, and deployment notes for internal tools, making onboarding easier for new developers and reducing repeated environment questions.
- Collaborated with senior developers on version control, code review, release preparation, and debugging practices that became the foundation for later AI/ML engineering work.

EDUCATION

Bachelor's degree in Computer Science

University of Amsterdam

2010 - 2014